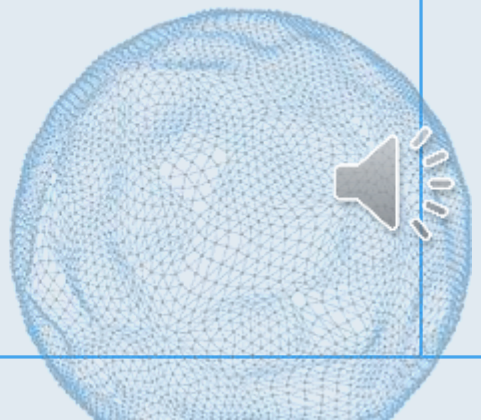
# Comparative Analysis Of Discriminative Vs. Generative Approaches For Med-VQA

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# PROBLEM MOTIVATION

- Medical images (X-ray, CT, MRI) contain complex visual patterns
- Med-VQA combines **vision + language**
- Important for decision support, education, and triage (with caution)



# METHODOLOGY – OVERVIEW

We compare **two fundamentally different approaches**:

**Discriminative baseline**

- CNN + Transformer classifier
- Predicts from a fixed answer vocabulary

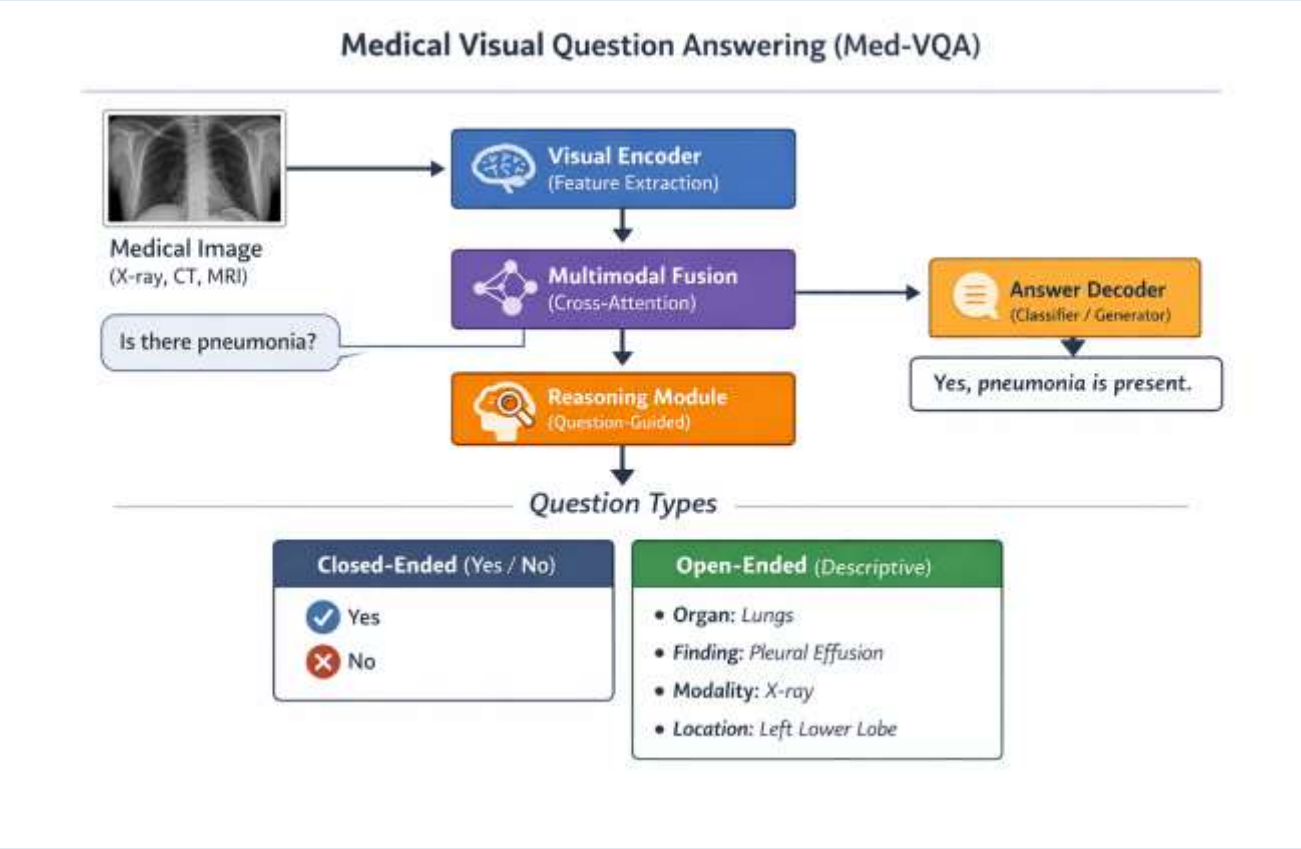
**Generative VLM**

- BLIP-2 + Flan-T5-XL
- Generates free-form answers

**Goal**: evaluate strengths and weaknesses for different question types



# METHODOLOGY – DATA & PIPELINE




**VQA-RAD** (open-source Med-VQA dataset)

Data Split

- Train: 1255
- Validation: 269
- Test: 269

# EXPERIMENTAL FAIRNESS & REPRODUCIBILITY



- Same dataset (VQA-RAD)
- Same closed vs open question definition
- Models evaluated on identical test samples



# DISCRIMINATIVE BASELINE MODEL

- We use a ResNet-50 encoder pre-trained on ImageNet to extract visual features.
- Simultaneously, a BERT-base model processes the text questions.
- These features are fused using element-wise multiplication and passed through a Multi-Layer Perceptron (MLP).



# BLIP-2 VISION–LANGUAGE MODEL



- **Model**: Salesforce/blip2-flan-t5-xl
- Trained for image-conditioned text generation
- **Fine-tuning method:**
- LoRA adapters applied to Q-Former attention layers
- Parameter-efficient adaptation

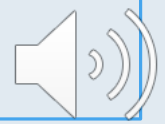
# EVALUATION METRICS

**Baseline (classification):**

- Accuracy
- Weighted F1
- Top-5 accuracy
- Closed vs open accuracy breakdown

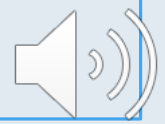
**BLIP-2 (generation):**

- Exact match (strict)
- BLEU
- ROUGE-L
- BERTScore-F1
- Token-level F1



# DISCRIMINATIVE PERFORMANCE

- Performance Metrics:
- Accuracy: 40.52%.
- Weighted F1 Score: 38.44%.
- Top-5 Accuracy: 59.48%.
- Summary: The baseline is robust for closed-set questions (e.g., "Yes/No") but is strictly limited by its "vocabulary bottleneck"—it cannot answer any question if the answer isn't in its fixed training dictionary.



# GENERATIVE PERFORMANCE

**Test Results**

Exact match: 0.4015

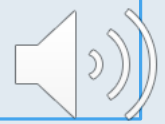
BLEU: 4.43

ROUGE-L: 0.4424

BERTScore-F1: 0.6115

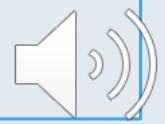
Token F1: 0.434

*"Performance diverges by question type: BLIP-2 maintains high semantic correctness on open-ended questions, while exact match drops due to phrasing variability."*



# HEAD-TO-HEAD COMPARISON

- Baseline and BLIP-2 achieve similar overall strict accuracy
- Baseline:
  - Strong on closed-ended yes/no questions
  - Limited by fixed answer vocabulary
- BLIP-2:
  - Better suited for open-ended Med-VQA
  - Can generate clinically plausible free-form answers



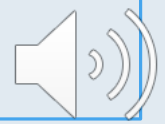
# LIMITATIONS & CRITICAL EVALUATION

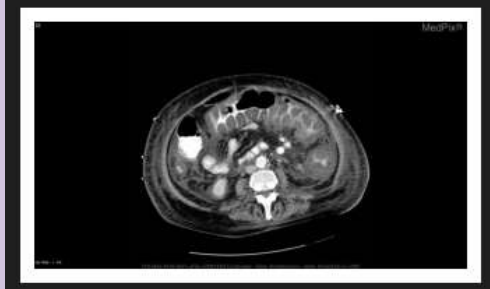
- Spatial Grounding Errors: Both models struggled with lateralization. We observed instances where the generative model correctly identified the organ but confused "left" and "right". This indicates that while the language model is strong, the vision-language alignment needs higher precision for radiology.
- Hallucinations: The generative model occasionally produced plausible-sounding but factually incorrect terms, such as diagnosing "aortic stenosis" when the visual evidence did not support it. This is a safety risk in clinical AI.
- Inference Cost: The generative model is computationally much heavier than the ResNet-BERT baseline, which poses a challenge for real-time deployment.
- Small Dataset Size



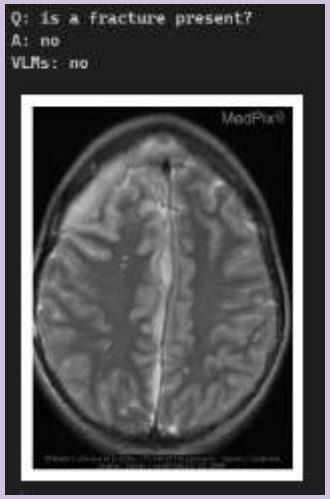
# CONCLUSION

- Conclusion: Generative Vision-Language Models (VLMs) are a promising direction for medical reasoning flexibility, especially for open-ended answers.
- Future Directions: Improving visual-spatial precision and increasing the diversity of low-resource medical training data.



| Discriminative Baseline Open Question | Discriminative Baseline Close Question |
| --- | --- |
| Q: does the image show pericolonic fat stranding?<br>A: yes<br>Baseline: yes<br>  | Q: what organ system is shown in the above image?<br>A: brain<br>Baseline: pineal gland<br>  |
| Q: is the mass heterogeneous?<br>A: yes<br>Baseline: yes<br>  | Q: how are the borders of the lesion?<br>A: smooth<br>Baseline: yes<br>  |

## Generative Close Question







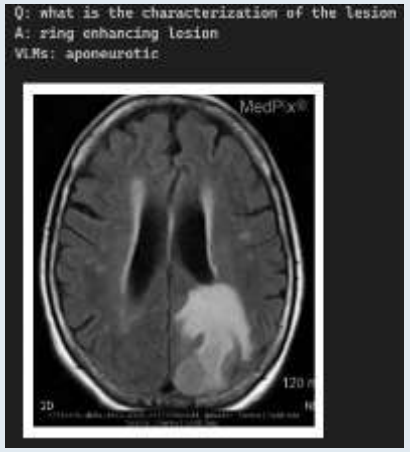


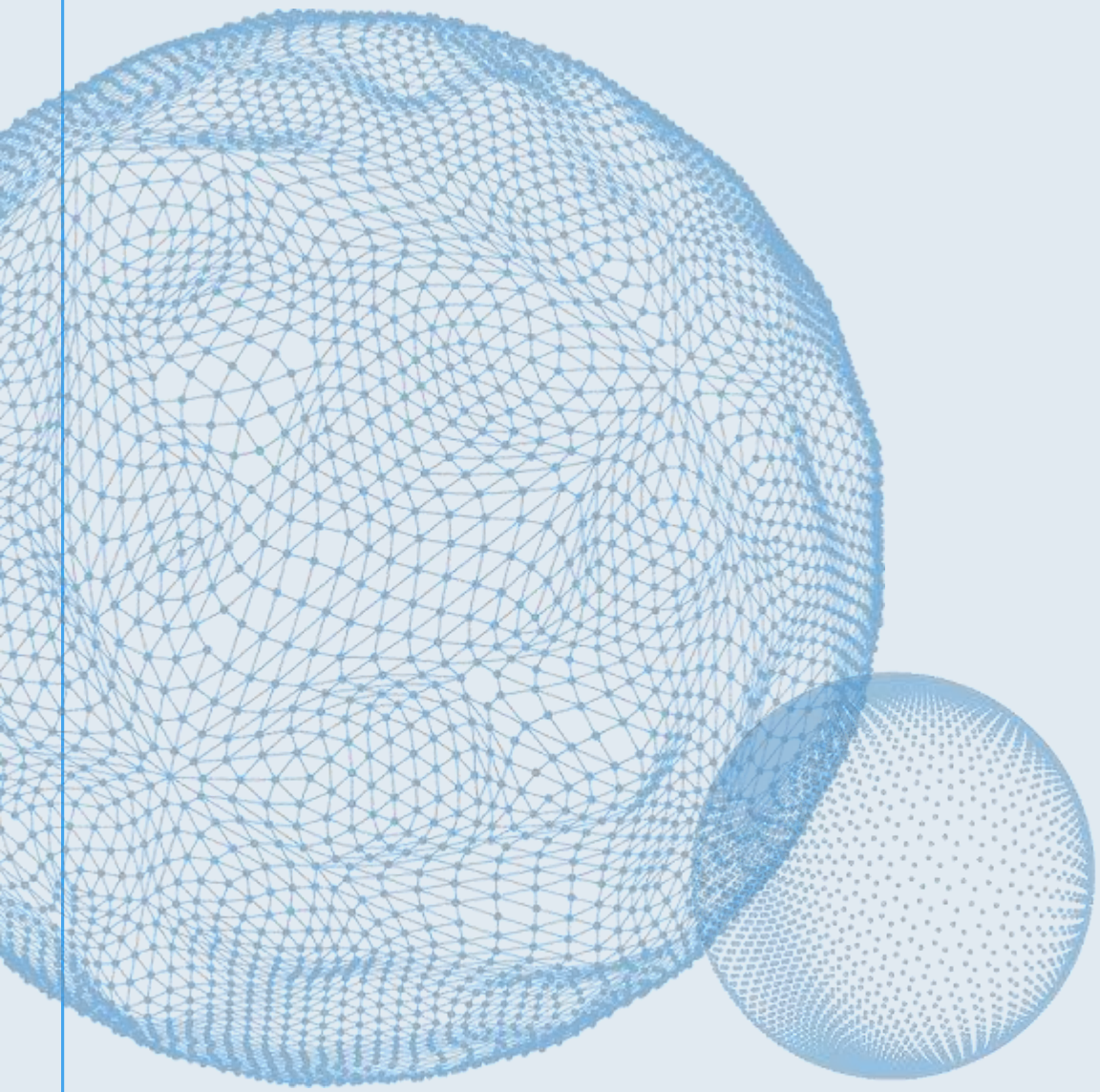

## Generative Open Question











# THANK YOU